

Technical Roadmap Palcos:

Overview:

The goal is to develop a product that allows owners of private boxes to rent their boxes for specific matches to the general public. This will allow palco owners to have additional income whenever they are not using the boxes, and allow the general public to experience a luxurious sporting experience without the need to own a private box. The Airbnb type platform where they rent and book the boxes is our product.

Technical breakdown:

Phase 1: Backend development:

- Need to set up an infrastructure to allow boxes to be rented by a large audience.
- Backend infrastructure will provide all functionality needed for owners *to* manage *and post* their private boxes and for renters to browse listings and complete and end *to* end booking.
- Future developments (not required for MVP) can include social aspects such as recommending listings to a friend, adding friends to reservation, having ranking of best renters, loyalty programs, etc.

Data Structure definitions:

Class object of a private box:

- Owner entity (1 person, uniquely tied to box)
 - Array of available dates to be rented
 - Array of dates for which a booking has been requested
 - Whenever a owner publishes their private box for the first time, a new object will be created. Each object should be unique and have an ID.
 - When they wish to publish their private box to be rented for a specific date, 'array of available dates' will be modified.
 - This should represent all the dates for which the private box can be rented.
 - Implement in method called by private box class named 'Add Listing()'
 - Dictionary containing a history of all past bookings and for each booking include:
 - Date of booking
 - Price rented
 - Price that owner received
 - Name of renter
 - Small description of event

Class object of a stadium:

- The stadium will be a collection of private boxes.
- At stadium level, should display/retrieve all boxes within the stadium available for rent.
 - Can store them in a sorted list by price/size/capacity, etc.

Class object of a user:

- Should contain basic information about a user such as name, email, payment information (can use 3rd party), location, stadium/team preferences, Instagram account, etc.
- Users should be able to look at available private boxes to rent based on their preferences or location.
- Need key method of rent(): allows users to create a request to rent a private box at the listed price. The owner will have to approve, then the user has to submit payment + deposit before reservation is finalized.

Booking Engine functionalities (actions renter object needs to be able to perform):

- Create renter account(): create basic profile for people who wish to rent a private box.
 - Have the option to connect to their social media (linkedin/Instagram) for owners to have more information when deciding whether to accept or decline a rent request.
 - Have the option to connect to a 3rd party payment platform so that they are able to successfully rent a box on their side. Can use stripe.
 - Screen 1 from UI will gather the information required for this API call.
 - Save preferences. This will help tailor the recommended boxes for each user. Preferences can include:
 - Price range
 - Preferred stadiums
 - Preferred teams and leagues
 - Expected capacity (buckets from 0-20, 20-50, etc).
 - Current location
- Create rent request(): whenever a user wishes to select a private box to rent, a 'request' object will be created with renter's information and box being requested to rent. The request object is sent to the booking engine.
 - Request information should include renting history of the user, email, age, income, social media, etc. as to provide the owner with sufficient information to evaluate the request of the user.
- Submit payment(): once the request of the user has been approved, prompt them in a new screen for them to submit the payment and deposit via a 3rd party payment app.

- Get instructions(): once a request has been approved, payment successfully processed, then send instructions to the user on what to expect during their reservation and how to access the box.
 - Can have instructions in a new screen of the app and/or sent via email.
- Post visit survey(): after the reservation has been completed, send a survey collecting information regarding their experience at the private box, in the booking process, and support along the way.
 - Will be key to collect this information for user retention.

Booking Engine Functionalities (actions system has to be able to perform):

- Process Rent Request(): listener object that receives booking requests created by users and then routes them to the appropriate private box owner.
 - If multiple requests are sent for the same box, then the engine will share all of them to the owner and the owner can accept whichever request he prefers. Solves the race condition problem.
- Accept booking(): once the owner selects which rent request to accept, the engine will mark the request as 'Accepted' and the user will be notified of next steps (payment).
- Reject booking(): if the owner decides not to rent their box to a specific user, then he will also be notified in a promptly manner so he can request similar boxes.
 - Have a standard message of a variety of reasons why this happens: another earlier request was accepted, the owner withdrew their listing, etc.
 - The same page also shows similar listings available so they can easily request a new one.
- Process payment(): once a request is accepted, the user will receive notification and be prompted to submit the payment deposit.
 - Deposit should be larger than rent price for any possible inconveniences.
 - Can be standard or decided by the owner.
 - The method 'submit payment()' from user object will connect to above 'Process Payment'
- Available boxes(): queries all private owner's boxes to see which dates they are available. Then sort by date and by stadium as variable 'Available boxes' to be used in renters feed.

Booking Engine functionalities (actions owner object needs to be able to perform):

- Create owner account(): create a profile for the owners with basic information. Will be required to enter the details of their private box to be able to open an owner account.
 - Once details are provided, the method will create an instance of the private box, link it to the owner and to a stadium.
 - After an account is created, the owner has the option to add a listing for their boxes. One owner can own multiple private boxes, all linked to the same account.
 - Owners should also connect to their preferred method of payment for the boxes they will be renting.
- Add listing (private box): when owners post a date for which they want to rent their private box.
 - Method goes into a private box object to modify an array of available dates.
 - New entry is created with price, date, capacity and short description.
 - Note that a listing is for one box for ONE specific date. Instead of one box being available to rent during any future date.
 - Can also provide owners with a feature to list their box to rent during any future date but will not be default behavior.
- Remove listing (private box): whenever owners want to remove a listing of any of their private boxes for a specific date. Will automatically remove from renters feed.

Feed functionality (for renter): will all rely on 'Available boxes' variable defined above and updated daily.

- Suggest stadium(): for renters only, based on their preferences, show a list of available boxes that fall within their suggested price range (+/-20%) and that most closely aligns with their preferred stadiums, team and location.
 - Can use k means approach with price proximity, location proximity, and team match as different metrics. Capacity can be included too if specified by user.
- Filter by location(): show all available boxes filtered by proximity. If boxes in the same location (stadium) then sort them by price from lowest to highest or by capacity.

- Filter by stadium(): show all available boxes for a given stadium. Within this screen can select to rank based on capacity and price.
- Show best deals(): show the list of private boxes currently being rented at a discount. The discount will be determined by the fair value of the box - renting price of box.
 - Possible reasons include last minute availability on behalf of owners, cancelled events, need of immediate cash flow, etc.
 - Fair value can be determined by an internal algorithm taking into account capacity, stadium, and location within the stadium.
- Key feature: location within stadium(): for each private box, need an internal map of the stadium to specify where the box is located (i.e. North east part of 5th floor, etc).

Phase 2: UI/Front end development:

Mobile app focused for users:

1. Screen 1: create account

- Basic screen asking for email, password, and link to social media.
- Will require at list one link to social media so that owners can have more information on renters when deciding which request to accept.
- Ask them to link their preferred payment method.
 - Need to decide if integrate with Apple Pay, Stripe or other 3rd party platform.

2. Screen 2: confirm account

- Basic step for users to confirm their account via email sent to them.

3. Screen 3: user preferences:

- Ask user for their preferences regarding:
 - Price range: how much they are willing to spend on renting one box for an event
 - Capacity: what capacity they are looking for when renting a box
 - Location: their current location to show available private boxes nearest to them
 - Preferred stadiums
 - Preferred teams: can show the private boxes available in the stadiums where their preferred teams are playing.

4. Screen 4: main landing page

- Will show the tailored list of suggested available private boxes based on their preferences
- Gives them one tab to filter private boxes by stadium

- Ranks stadiums by proximity
 - Within stadium can also filter by price or capacity
- Gives them another tab (map tab) to see the different stadiums and how far they are from the user's location.

5. Screen 5: submit payment

- After a request has been sent by the renter, and accepted by the owner, the booking engine will notify the user session to submit a payment.
- This will prompt users to a new screen where they will see the finalized amount due (including booking fee + deposit) to submit the payment.
- Notify that the money will go into escrow until the booking is complete.
- If there are no issues found at the private box, then escrow will automatically return the money to the user's account.

6. Screen 6: Receive confirmation:

- Once payment has been successfully processed, the user will receive a notification.
- This will include reservation details, confirmation code, instructions on how to enter the private box, and what to expect during the day of reservation.
- The same information will be sent via email.
- Need to generate a valid stadium entry ticket at this stage.

7. Screen 7: After event survey:

- Quick survey asking users about their overall experience both within the box itself and during the booking and payment process.
- Key to improving user experience on our side.